KALYANI GOVERNMENT ENGINEERING COLLEGE

Kalyani, Nadia, West Bengal – 741235

(Affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal)

DEPARTMENT OF COMPUTER APPLICATION

A PROJECT REPORT ON

"T-TEST SIMULATOR FOR HYPOTHESIS TESTING"

An Interactive Web Application for One-Sample, Independent and Paired t-Tests

Submitted in partial fulfilment of the requirements of the subject
Basic Data Science (MCANE304F)
of the programme **Master of Computer Applications (MCA)**

Submitted by

Sl.	Student Name	Roll No.
1	Sarbajit Chaki	10271025033
2	Somnath Ganguly	10271025040
3	Sanjib Kumar Dera	10271025030

Under the Guidance of
Dr. Anuradha Banerjee
Project Guide, Department of Computer Application

Live application: https://frontend-kappa-steel-41.vercel.app/

November 2026

CERTIFICATE

This is to certify that the project report entitled **"T-Test Simulator for Hypothesis Testing"**, submitted by **Sarbajit Chaki** (Roll No. 10271025033), **Somnath Ganguly** (Roll No. 10271025040) and **Sanjib Kumar Dera** (Roll No. 10271025030), students of the **Master of Computer Applications (MCA)** programme, Department of Computer Application, Kalyani Government Engineering College, in partial fulfilment of the requirements of the subject **Basic Data Science (MCANE304F)**, is a bona fide record of the work carried out by them under my supervision and guidance.

To the best of my knowledge, the work embodied in this report is original and has not been submitted elsewhere for any other degree, diploma or course.

Dr. Anuradha Banerjee
Project Guide
Department of Computer Application
Kalyani Government Engineering College

_____
Head of the Department
Department of Computer Application
Kalyani Government Engineering College

Date: _____

Place: Kalyani

DECLARATION

We, the undersigned students of the Master of Computer Applications (MCA) programme, Department of Computer Application, Kalyani Government Engineering College, hereby declare that the project report entitled **"T-Test Simulator for Hypothesis Testing"** is the result of our own work carried out under the guidance of **Dr. Anuradha Banerjee**.

We further declare that this work is original, that all sources of information used have been duly acknowledged in the references, and that it has not been submitted in part or in full to any other institution or for any other course.

_____	_____	_____
Sarbajit Chaki	**Somnath Ganguly**	**Sanjib Kumar Dera**
Roll No. 10271025033	Roll No. 10271025040	Roll No. 10271025030

Date: _____

Place: Kalyani

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Sarbajit Chaki (10271025033)
Somnath Ganguly (10271025040)
Sanjib Kumar Dera (10271025030)

ABSTRACT

Hypothesis testing is a core technique of statistics and data science, and the Student's t-test is its most common form. Learners, however, find it hard to choose the correct variant of the test, to select the direction of the alternative hypothesis, to read the output and to check the assumptions behind it. This project presents the **T-Test Simulator for Hypothesis Testing**, an interactive web application that performs the one-sample, independent two-sample (Welch and pooled) and paired t-tests.

The user selects a test, supplies data by generation with a reproducible random seed, by uploading a CSV file or by typing values, sets the alternative hypothesis (two-sided, left-tailed or right-tailed) and the significance level (0.01, 0.05 or 0.10), and presses a single button. The application returns the t-statistic, degrees of freedom, p-value, confidence interval, standard error and Cohen's d, announces the decision on the null hypothesis, and writes a plain-language interpretation with warnings for small samples and non-normality. It displays the t-distribution with its rejection regions, a histogram, group means with error bars, Shapiro-Wilk and Levene diagnostics and a log of the last ten runs, and exports results and data as CSV files.

The application was tested on three sample datasets (exam scores, a drug trial and a weight-loss study) in all nine combinations of test and tail. The results were consistent with theory, and key values were confirmed by independent hand calculation. The report describes the theory, analysis, design, usage, testing and validation of the system, together with its limitations and future scope.

Keywords: hypothesis testing, t-test, Welch's test, paired t-test, p-value, confidence interval, Cohen's d, web application, data science.

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CHAPTER 1: INTRODUCTION

1.1 Background

Hypothesis testing is a core technique of statistics and data science: it decides whether a difference seen in a *sample* is large enough to be believed about the *population*, or could be due to chance. The Student's t-test, introduced by William Sealy Gosset ("Student") in 1908 [1], is its most widely used form. Three variants cover most situations: the **one-sample** test (one mean against a known value), the **independent two-sample** test (two unrelated groups) and the **paired** test (two measurements on the same subjects).

1.2 Problem Statement

Learners find t-tests difficult because the arithmetic is slow and needs critical-value tables; professional packages (SPSS, R, Python) need installation or programming skills; most tools give bare numbers without explaining the decision; the effect of a left-, right- or two-tailed alternative is hard to see; and assumption checks are often skipped. A simple, freely accessible and explanatory tool is therefore needed.

1.3 Objectives

- To build a web application performing one-sample, independent (Welch and pooled) and paired t-tests.
- To accept generated (seeded), uploaded CSV and manually typed data, with two-, right- and left-tailed hypotheses at $\alpha = 0.01$, 0.05 and 0.10.
- To report t, df, p-value, confidence interval, standard error and Cohen's d, with a decision, plain-language interpretation and warnings.
- To visualise results (t-curve with rejection regions, histograms, group means, diagnostics) and to validate all nine test/tail combinations.

1.4 Scope and Organisation

The project covers the three t-tests, assumption diagnostics (Shapiro-Wilk, Levene), automatic interpretation, a history of the last ten runs and CSV export. ANOVA, non-parametric tests, power analysis and user accounts are outside scope (Chapter 7). Chapter 2 gives the theory, Chapter 3 the analysis, Chapter 4 the design, Chapter 5 the inputs, outputs and usage, and Chapter 6 the testing and validation.

CHAPTER 2: THEORETICAL BACKGROUND

2.1 Hypothesis Testing and Errors

The **null hypothesis (H_0)** states no effect or difference; the **alternative (H_1)** states what is being looked for. The **p-value** is the probability of a result at least as extreme as observed if H_0 were true. With **significance level α** chosen in advance, H_0 is rejected when $p < \alpha$;

otherwise we *fail to reject* H_0, which is not proof that H_0 is true [8]. α is the risk of a Type I error (false rejection); β is the risk of a Type II error (missing a real effect). In the application α = 0.01, 0.05 or 0.10, which also sets the confidence level to 99%, 95% or 90%.

2.2 The t-Distribution and the Three Tests

When the population standard deviation is unknown, the standardised mean follows Student's t-distribution with *df* degrees of freedom: bell-shaped like the normal curve but with heavier tails, approaching normal as df grows. For a sample of size n with mean $\bar{x}$ and standard deviation s, the **one-sample** statistic is:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \qquad df = n - 1, \qquad SE = \frac{s}{\sqrt{n}}$$

The **independent two-sample** test uses Welch's statistic by default (no equal-variance assumption), with df from the Welch–Satterthwaite approximation [2]:

$$t = \frac{\bar{x}_1 - \bar{x}_2 - \Delta_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Ticking *Assume Equal Variances* gives Student's pooled test:

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}, \qquad t = \frac{\bar{x}_1 - \bar{x}_2 - \Delta_0}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}, \qquad df = n_1 + n_2 - 2$$

The **paired** test forms differences $d_i = A_i - B_i$ and applies a one-sample test to them:

$$t = \frac{\bar{d} - \Delta_0}{s_d/\sqrt{n}}, \qquad df = n - 1$$

2.3 Tails, Confidence Intervals and Effect Size

A two-sided alternative (≠) splits α between both tails; right-tailed (>) and left-tailed (<) alternatives put all of α in one tail (Figure 2.1). A one-tailed test rejects with a smaller t in its chosen direction but cannot detect an effect in the opposite direction, so the tail must be chosen before seeing the data.

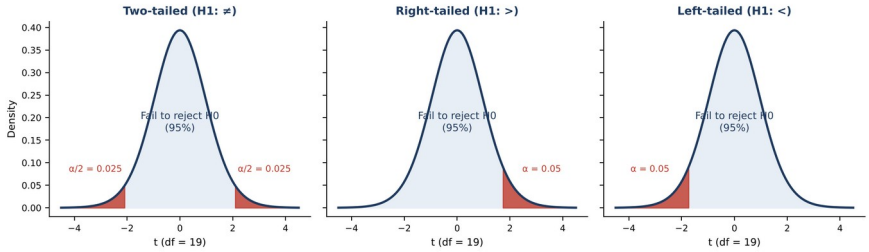


Figure 2.1: Rejection regions of the t-distribution (df = 19, α = 0.05)

The two-sided confidence interval is estimate ± t(α/2, df) × SE; in one-sided tests one end is shown as −∞ or +∞. **Cohen's d** measures practical size in standard-deviation units [5]: $(\bar{x} - \mu_0)/s$ for one sample, $(\bar{x}_1 - \bar{x}_2)/s_P$ for independent samples and $\bar{d}/s_d$ for paired samples. The application labels |d| below 0.2 negligible, 0.2–0.5 small, 0.5–0.8 medium and 0.8 or more large.

2.4 Assumptions and Diagnostics

The tests assume independent observations, approximate normality (of the data or paired differences) and, for the pooled test, equal variances. The application reports the **Shapiro-Wilk** test for normality [3] and **Levene's** test for equal variances [4], and shows an amber warning when the sample size is below 15, because with small samples the normality assumption is critical.

CHAPTER 3: SYSTEM ANALYSIS

3.1 Existing Approaches and Proposed System

Existing ways to run a t-test are manual calculation with tables (slow, error-prone), spreadsheet functions (bare numbers, easy to pick the wrong tail), statistical packages (installation, licences or coding skills) and generic online calculators (usually one test only, little explanation). The proposed system is a single web application that combines data input, computation, visualisation and explanation for all three t-tests behind one consistent interface. It is feasible technically (standard statistics, modern browsers), economically (free-tier hosting, no licensed software) and operationally (no training beyond basic statistics).

3.2 Requirements

Table 3.1: Functional requirements

ID	Requirement
FR-1	Select One-Sample, Independent 2-Sample or Paired Samples t-Test.
FR-2	Supply data by generation (size, mean, SD, distribution, seed), CSV upload (choose columns) or manual entry.
FR-3	Choose H_1 (two-sided, left-, right-tailed), α (0.01, 0.05, 0.10), μ_0 or Δ_0, and Welch or pooled variance.
FR-4	Compute t, df, p-value, mean difference, confidence interval, SE and Cohen's d.
FR-5	Show the decision with H_0 and H_1, a written interpretation and small-sample / non-normality warnings.
FR-6	Draw the t-curve, histogram, group means with error bars; report Shapiro-Wilk and Levene diagnostics.
FR-7	Keep a log of the last ten runs; export results and data as CSV.

Non-functional requirements: simple single-page usability, numerical accuracy to the displayed precision, reproducibility (same seed gives same data), immediate response, and portability across current browsers (Chrome, Edge, Firefox, Safari) on desktop and mobile with only an internet connection.

CHAPTER 4: SYSTEM DESIGN

4.1 Architecture and Modules

The application uses three layers (Figure 4.1): a presentation layer (test selector, configuration panel, result cards, decision banner, charts), a processing layer (data validation, t-test computation, Shapiro-Wilk/Levene diagnostics, interpretation generator) and a data and output layer (generated, uploaded or typed data; run history; CSV export).

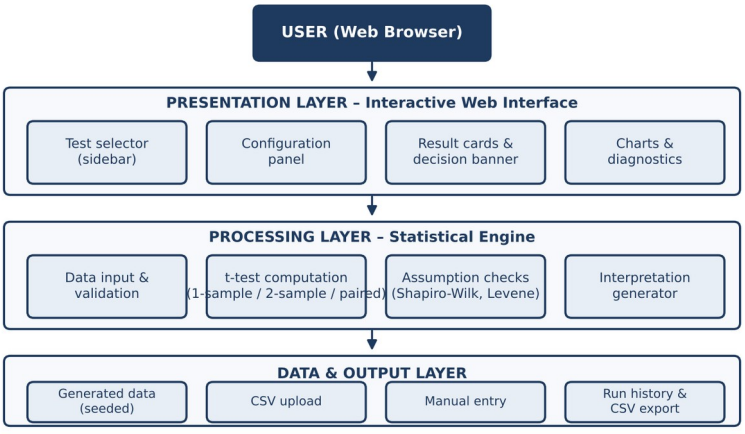


Figure 4.1: Layered architecture of the T-Test Simulator

4.2 Data Flow and User Interface

Figure 4.2 shows the level-0 data flow: the user supplies data, the test type and hypothesis settings and receives the results, decision, charts and warnings; each run is logged and can be exported. The workspace, from top to bottom, holds the configuration panel, four result cards, the decision banner with hypotheses and interpretation, the charts, the diagnostics and the run history; a sidebar selects the test.

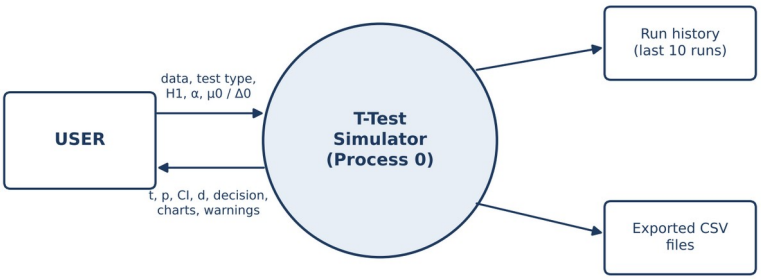


Figure 4.2: Level-0 data flow diagram

Design choices that aid learning: a green banner for "Reject H_0", a tag showing whether $p < \alpha$, the observed t drawn against the red rejection regions, and the p-value shown as "< 0.0001" when tiny. The difference is always Sample A minus Sample B, and groups are labelled Sample A and Sample B.

CHAPTER 5: IMPLEMENTATION AND USER GUIDE

5.1 Deployment and Access

The T-Test Simulator is a web application deployed on Vercel and opened in any browser without installation or login: **https://frontend-kappa-steel-41.vercel.app/**. Its page title is *Interactive T-Test Simulator | Scientific Analytics Studio*.

5.2 Inputs and Outputs

Table 5.1: Inputs

Input	Choices / action
Data source	**Generate** (size, target mean, SD, distribution shape, seed, pair correlation for paired); **Upload CSV** (pick numeric columns for Sample A / B); **Manual Entry** (numbers separated by commas, spaces or new lines)
Alternative hypothesis (H_1)	Two-Sided, Left-Tailed, Right-Tailed
Significance level (α)	0.01, 0.05, 0.10 (also sets 99%, 95%, 90% confidence level)
μ_0 / Δ_0	Hypothesised mean (one-sample) / difference (independent, paired; normally 0)
Assume equal variances	Independent test only: off = Welch (default), on = Student's pooled

Table 5.2: Outputs

Output	How to read it
t-Statistic and df	Standard errors between the result and what H_0 predicts, and degrees of freedom.
p-Value	Probability of a result at least this extreme if H_0 were true; tagged significant when $p < \alpha$.
Mean Diff & CI	Observed difference and its confidence interval (one end $-\infty/+\infty$ in one-tailed tests). For the one-sample test the interval is for the population mean itself.
Cohen's d and SE	Effect size label and standard error.
Decision banner	"Reject H_0" if $p < \alpha$, otherwise "Fail to reject H_0", with H_0 and H_1 written out.
Interpretation, charts, diagnostics	Plain-language reading and warnings; t-curve with rejection regions, histogram, group means, descriptive table; Shapiro-Wilk and Levene p-values.

5.3 Using the Application

1. Open the URL and choose the test in the sidebar.
2. Choose Generate, Upload CSV or Manual Entry and supply the data (select the CSV column(s) for Sample A / B).
3. Set the alternative hypothesis, α, μ_0 or Δ_0 and (independent test) the equal-variance option.
4. Press **Run T-Test Analysis**, read the cards, banner, charts, diagnostics and any amber warning, and export CSV files if needed.

5.4 Sample Datasets

Table 5.3: Sample datasets used for testing

File and column	Test	n	Mean	Std. dev.
one_sample_exam_scores.csv: score	One-sample, $\mu_0 = 75$	20	84.28	5.57
two_sample_drug_trial.csv: placebo	Independent, Sample A	15	122.67	4.06
two_sample_drug_trial.csv: treatment	Independent, Sample B	15	111.67	4.25
paired_weight_loss.csv: before_diet	Paired, Sample A	12	86.48	6.10
paired_weight_loss.csv: after_diet	Paired, Sample B	12	83.62	5.50

CHAPTER 6: TESTING, RESULTS AND VALIDATION

6.1 Strategy and Test Cases

Black-box functional testing was done in the application itself: each test was run on its sample dataset (loaded through Upload CSV) with all three tail settings, at $\alpha = 0.05$ and $\mu_0 = 75$ or $\Delta_0 = 0$, using Welch's test unless stated. Three further cases covered the pooled test, the diagnostics and the small-sample warning.

Table 6.1: Test cases

ID	Test / tail	Expected	Observed	Result
TC-01	One-sample, two-tailed	Reject H_0 (mean 84.28 ≫ 75)	t = 7.4515, df = 19, p < 0.0001; Reject	Pass
TC-02	One-sample, right-tailed	Reject H_0	CI (82.12, +∞); Reject	Pass
TC-03	One-sample, left-tailed	Fail to reject	p = 1.0000; CI (−∞, 86.43); Fail to reject	Pass
TC-04	Independent, two-tailed	Reject H_0	t = 7.2407, df = 27.94, p < 0.0001; Reject	Pass
TC-05	Independent, right-tailed	Reject H_0	CI (8.42, +∞); Reject	Pass
TC-06	Independent, left-tailed	Fail to reject	p = 1.0000; CI (−∞, 13.58); Fail to reject	Pass
TC-07	Paired, two-tailed	Reject H_0	t = 15.0225, df = 11, p < 0.0001; Reject	Pass
TC-08	Paired, right-tailed	Reject H_0	CI (2.52, +∞); Reject	Pass
TC-09	Paired, left-tailed	Fail to reject	p = 1.0000; CI (−∞, 3.21); Fail to reject	Pass
TC-10	Independent, pooled	Same decision as Welch	t = 7.2407, df = 28; same decision	Pass
TC-11	Diagnostics	Normality / variance reported	Shapiro-Wilk 0.9171; 0.9996, 0.9316; Levene 0.8237	Pass
TC-12	Paired, n = 12	Amber warning, run completes	Warning shown; results displayed	Pass

6.2 Results

One-sample (exam scores). The sample mean is 84.28, i.e. 9.28 above μ_0 = 75. The two-tailed result is t = 7.4515 (df = 19), p < 0.0001, 95% CI [81.67, 86.88] and Cohen's d = 1.67 (Large): Reject H_0 (Figure 6.1). The t-curve rejection regions begin at about ±2.093 and the normality p-value is 0.9171, so no warning appears (Figure 6.2). The left-tailed run gives p = 1.0000 because the scores lie on the opposite side of 75.

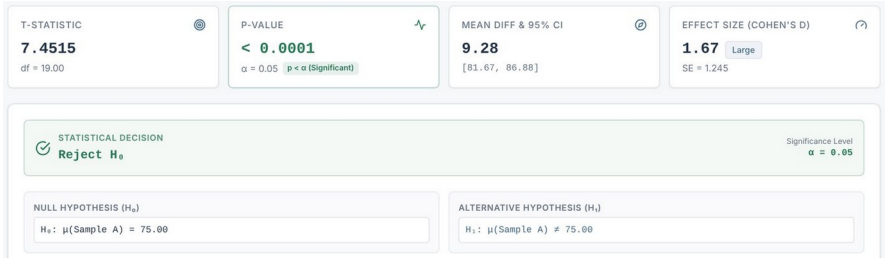


Figure 6.1: One-sample test – two-tailed result cards, decision and hypotheses

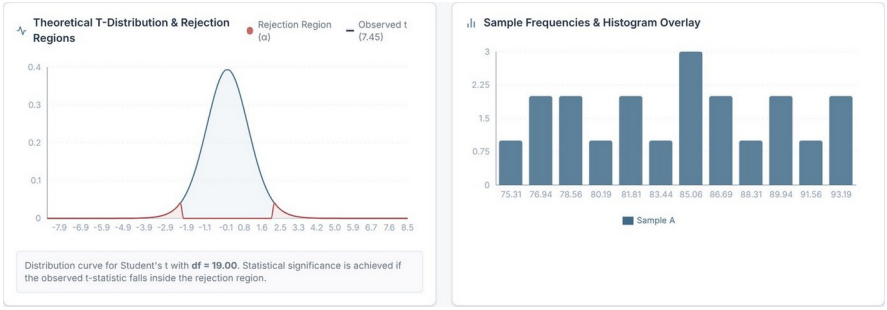


Figure 6.2: One-sample test – t-distribution and histogram

Independent (drug trial, Welch). The means are 122.67 (placebo) and 111.67 (treatment), a difference of 11.00 with SE 1.519. The result is t = 7.2407, Welch df = 27.94, p < 0.0001,

CI [7.89, 14.11], d = 2.64 (Large): Reject H_0 (Figure 6.3). Shapiro-Wilk p-values are 0.9996 and 0.9316 and Levene's p is 0.8237, all passes. The right-tailed run rejects H_0 and the left-tailed run fails to reject, because placebo is the higher group (Figure 6.4). The difference is always A − B, so swapping columns swaps these outcomes. The pooled test gives the same t with df = 28 because spreads and sizes are almost equal.

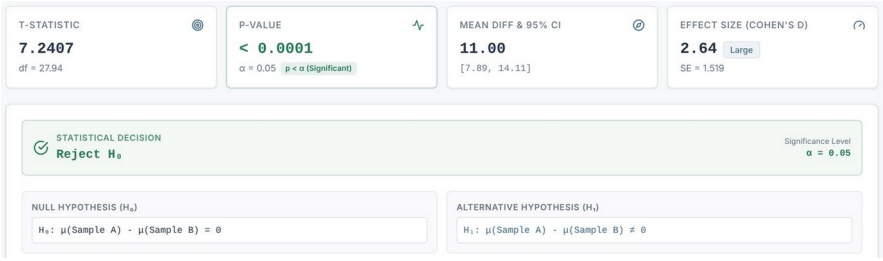


Figure 6.3: Independent test – two-tailed result cards, decision and hypotheses

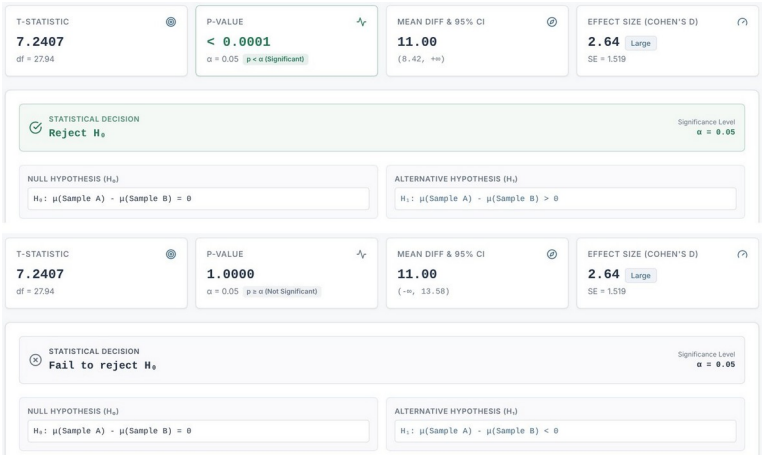


Figure 6.4: Independent test – right-tailed (top) and left-tailed (bottom) results

Paired (weight loss). Differences (before − after) average 2.87 kg with SE 0.191, giving t = 15.0225 (df = 11), p < 0.0001, CI [2.45, 3.29] and d = 4.34 (Large): Reject H_0 (Figure 6.5). The right-tailed run (before > after) rejects H_0 with CI (2.52, +∞); the left-tailed run gives p = 1.0000 because nobody gained weight. An amber box warns that n = 12 is below 15 and normality is critical; it is a caution, not an error (Figure 6.6).

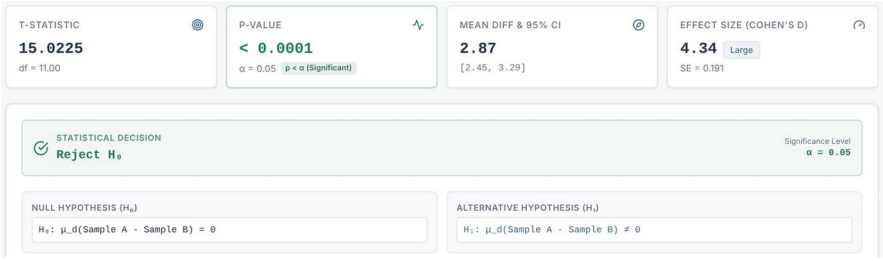


Figure 6.5: Paired test – two-tailed result cards, decision and hypotheses

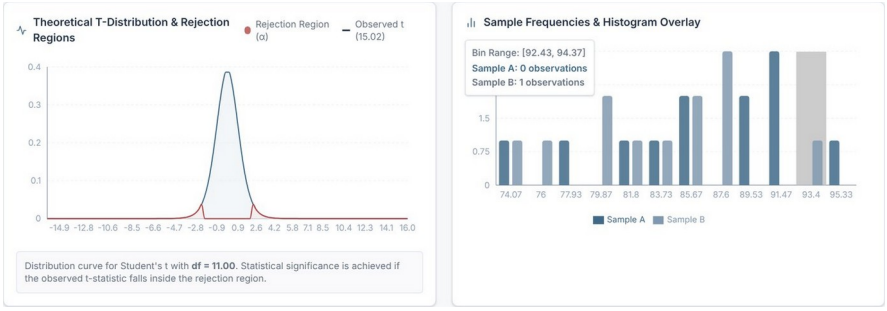


Figure 6.6: Paired test – t-distribution and histogram of both measurements

Table 6.2: Summary of the nine runs

Test	Tail	t	df	p shown	Decision
One-sample	Two / Right / Left	7.4515	19	< 0.0001 / < 0.0001 / 1.0000	Reject / Reject / Fail
Independent	Two / Right / Left	7.2407	27.94	< 0.0001 / < 0.0001 / 1.0000	Reject / Reject / Fail
Paired	Two / Right / Left	15.0225	11	< 0.0001 / < 0.0001 / 1.0000	Reject / Reject / Fail

The t-statistic is identical for the three tails of a test; only the p-value, interval, rejection region and decision change.

6.3 Validation Against Hand Calculation

Key results were recomputed from the summary statistics in Table 5.3 using the Chapter 2 formulas and SciPy [9] for t quantiles. Agreement is expected to within rounding of the two-decimal inputs.

Table 6.3: Reported values versus independent recomputation

Quantity	Reported	Recomputed	Agreement
One-sample t (SE = 5.57/√20 = 1.2455)	7.4515	7.451	Yes
One-sample 95% CI; critical t (df 19)	[81.67, 86.88]; ≈ 2.093	[81.67, 86.89]; 2.0930	Rounding only
One-sample Cohen’s d	1.67	1.666	Yes
Independent SE; t; Welch df	1.519; 7.2407; 27.94	1.5176; 7.248; 27.94	Rounding only
Independent 95% CI; right-tailed bound	[7.89, 14.11]; 8.42	[7.89, 14.11]; 8.42	Yes
Paired t = 2.87/0.191; critical t (df 11)	15.0225; ≈ 2.201	15.03; 2.2010	Yes
Paired 95% CI; Cohen’s d	[2.45, 3.29]; 4.34	[2.45, 3.29]; 4.34	Yes

All twelve test cases behaved as expected. The left-tailed runs show the main risk of a one-sided test: when the effect lies in the opposite direction, p is 1.0000 however large the difference. The large effect sizes (d from 1.67 to 4.34) reflect the strong effects built into the sample datasets; real data usually give smaller values.

CHAPTER 7: ADVANTAGES, LIMITATIONS AND FUTURE SCOPE

Advantages. One interface for three t-tests; no installation or licence; three ways to supply data with reproducible seeded generation; the t-curve links the numerical decision to its geometric meaning; plain-language interpretation, effect size and diagnostics encourage correct reasoning; warnings never block the analysis; CSV export and run history.

Limitations. Only t-tests are supported (no ANOVA or non-parametric tests); groups cannot be named (Sample A / Sample B); only the last ten runs are kept; no power analysis; for badly skewed small samples the application only warns and offers no alternative test.

Future scope. Non-parametric alternatives (Wilcoxon, Mann-Whitney) suggested when normality fails; one-way ANOVA with post-hoc tests; power and sample-size calculator; custom group names and units; chart and PDF export; user accounts with saved datasets; and a "which test should I use?" helper.

CHAPTER 8: CONCLUSION

The project delivers an interactive web application that performs the one-sample, independent and paired t-tests, reports t, df, p-value, confidence interval, standard error and Cohen's d, announces the decision, explains it in plain language and checks the assumptions. Tested on three sample datasets in all nine test/tail combinations, it produced consistent results: exam scores significantly above 75, placebo readings significantly higher than treatment readings, and a significant average diet loss of about 2.87 kg. Independent recomputation agreed with the reported values to within rounding.

The work also reinforces good practice: choose the test from how the data were collected, choose the tail from the research question before seeing the data, read the p-value together with the confidence interval and effect size, and check the diagnostics before trusting the conclusion.

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